

Overview: The Skin as a Window to Systemic Health

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2025 Edition

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Chapter 1 Overview: The Skin as a Window to Systemic Health

Overview: The Skin as a Window to Systemic Health

The skin is far more than a passive covering — it is the largest organ of the human body, comprising approximately **15% of total body weight** and covering a surface area of roughly **1.5 to 2 m²** in adults. It serves as an essential living biological barrier against mechanical injury, ultraviolet radiation, microbial invasion, and chemical irritants. Beyond its protective role, the skin is a dynamic organ that regulates body temperature through vasodilation and vasoconstriction of dermal blood vessels, controls water loss via the stratum corneum, participates in immune surveillance through Langerhans cells and dermal dendritic cells, and synthesizes vitamin D upon exposure to ultraviolet B radiation.

Critically, the skin frequently serves as a mirror of underlying systemic disease. Changes on the skin surface may be the first — and sometimes the only — herald of serious internal pathology, including malignancy, autoimmune disease, endocrine dysfunction, infection, and metabolic derangement. A thorough dermatological examination, encompassing the entire skin surface, nails, hair, and mucous membranes, should therefore be considered an integral component of every comprehensive clinical assessment.

The psychological burden of skin disease must not be underestimated. Visible skin conditions carry profound psychosocial consequences, including social stigmatization, reduced self-esteem, depression, anxiety, and impaired occupational and interpersonal functioning. The **Dermatology Life Quality Index (DLQI)** is a validated, widely used, 10-item questionnaire that quantifies the impact of skin disease on a patient's quality of life over the preceding week. Scores range from 0 (no impact) to 30 (extreme impact), with established interpretive bands: 0–1 (no effect), 2–5 (small effect), 6–10 (moderate effect), 11–20 (very large effect), and 21–30 (extremely large effect).

KEY TEACHING POINTS

Skin is the largest organ of the body (~15% body weight, 1.5–2 m² surface area in adults).

The skin provides a biological barrier, regulates temperature, prevents water loss, and participates in immune surveillance and vitamin D synthesis.

Underlying systemic disease can be heralded by changes on the skin surface — always examine the whole skin, nails, hair, and mucous membranes.

The DLQI is a validated 10-item questionnaire (score 0–30) measuring the impact of skin disease on quality of life.

DLQI interpretive bands: 0–1 (no effect), 2–5 (small), 6–10 (moderate), 11–20 (very large), 21–30 (extremely large).

MCQ TRAPS & PITFALLS

Do NOT assume that a rash is purely dermatological — always consider systemic causes, especially when constitutional symptoms are present.

DLQI score ranges are frequently tested — memorize the exact bands and their corresponding categories.

Skin Anatomy: The Three Layers

1. Epidermis

The epidermis is the outermost layer of the skin and is composed predominantly of **keratinocytes** — stratified squamous epithelial cells that undergo terminal differentiation as they migrate from the basal layer

to the surface. The epidermis is **avascular**, receiving its nutritional supply by diffusion from the underlying dermal capillaries. It is rich in **keratin**, the structural protein that provides mechanical resilience.

The epidermis is organized into five distinct strata, from deepest to most superficial:

- **Stratum basale (basal layer):** A single layer of columnar to cuboidal cells resting on the basement membrane zone. Contains mitotically active stem cells responsible for continuous epidermal renewal. Melanocytes are interspersed among basal keratinocytes at a ratio of approximately 1:10. Merkel cells (mechanoreceptors for light touch) are also found here.
- **Stratum spinosum (prickle cell layer):** Several layers of polyhedral keratinocytes connected by desmosomes (giving a "spiny" appearance on histology). Langerhans cells — the principal antigen-presenting cells of the epidermis — are most abundant in this layer.
- **Stratum granulosum (granular layer):** Three to five layers of flattened keratinocytes containing keratohyalin granules (profilaggrin, loricrin) and lamellar bodies (which extrude lipid contents into the intercellular space to form the permeability barrier).
- **Stratum lucidum:** A thin, translucent layer found only in thick skin (palms and soles), consisting of densely packed dead keratinocytes.
- **Stratum corneum (horny layer):** The outermost layer, composed of 15–20 layers of anucleate, flattened corneocytes embedded in a lipid matrix. This is the primary physical and permeability barrier of the skin.

The entire process of epidermal turnover — from basal cell division to shedding of corneocytes — takes approximately **28–30 days** in normal skin. This turnover rate is accelerated in psoriasis (reduced to 3–5 days).

2. Dermis

The dermis is the connective tissue layer situated beneath the epidermis, separated by the **basement membrane zone** (dermal-epidermal junction). It provides structural support, elasticity, and nutritional supply to the avascular epidermis. The dermis contains:

- **Connective tissue:** Collagen fibers (primarily type I and III) provide tensile strength; elastin fibers provide elasticity and resilience. These are embedded in a ground substance of glycosaminoglycans (hyaluronic acid, dermatan sulfate) and proteoglycans.
- **Blood vessels:** A superficial vascular plexus (subpapillary plexus) and a deep vascular plexus (reticular dermis) regulate thermoregulation and nutrient delivery.
- **Lymphatic vessels:** Drain interstitial fluid and transport immune cells.
- **Nerves:** Both sensory (free nerve endings, Meissner corpuscles for light touch, Pacinian corpuscles for deep pressure/vibration) and autonomic nerves (supplying blood vessels, arrector pili muscles, eccrine glands).
- **Skin appendages:** Hair follicles, sebaceous glands, eccrine sweat glands, and apocrine sweat glands.

The dermis is divided into the **papillary dermis** (superficial, thin, interdigitating with epidermal rete ridges) and the **reticular dermis** (deeper, thick, containing dense irregular connective tissue).

3. Subcutis (Hypodermis)

The subcutis is the deepest layer of the skin, composed primarily of **adipose tissue** (fat lobules separated by fibrous septa). It serves as:

- An energy reservoir (fat storage and metabolism)

- Thermal insulation
- Mechanical cushioning and shock absorption
- A conduit for larger blood vessels and nerves

The thickness of the subcutis varies significantly with anatomical site, nutritional status, age, sex, and hormonal influences.

KEY TEACHING POINTS

The epidermis is avascular and receives nutrients by diffusion from dermal capillaries.

Epidermal turnover takes ~28–30 days; this is reduced to 3–5 days in psoriasis.

Five epidermal strata (deepest to most superficial): basale, spinosum, granulosum, lucidum (thick skin only), corneum.

The basement membrane zone separates the epidermis from the dermis and is the site of blistering in diseases such as bullous pemphigoid.

The subcutis is the deepest layer, composed primarily of adipose tissue providing insulation and cushioning.

Red Flag Clinical Features on Skin Examination

When examining the skin, certain clinical features should immediately raise concern for serious underlying pathology. These "red flag" findings demand thorough investigation and, in many cases, urgent referral.

1. Non-Blanching (Palpable) Purpura

Purpuric lesions that do not blanch on diascopy (pressure with a glass slide) indicate **extravasation of red blood cells** from the vasculature into the surrounding tissue. This is a cardinal sign of:

- **Vasculitis** (inflammatory destruction of blood vessel walls) — e.g., IgA vasculitis (Henoch-Schönlein purpura), leukocytoclastic vasculitis
- **Sepsis/meningococcemia** (meningococcal septicemia classically causes purpurae that coalesce into widespread purpura fulminans)
- **Thrombocytopenia or coagulopathy** (due to bleeding diathesis)

Diascopy technique: Press a glass slide firmly against a skin lesion. Blanching lesions (erythema, vasodilation) will pale. Non-blanching lesions (purpura, petechiae, vasculitis) will retain their red/purple color because red blood cells have already extravasated into the tissue.

2. Constitutional Symptoms Accompanying Skin Lesions

The presence of systemic symptoms — fever, weight loss, night sweats, malaise, arthralgias — in association with skin findings should raise suspicion for:

- Underlying malignancy (particularly lymphoma, leukemia)
- Autoimmune/connective tissue disease (SLE, dermatomyositis)
- Serious infection (disseminated fungal, mycobacterial, viral)
- Drug reaction (DRESS syndrome, Stevens-Johnson syndrome)

3. Rapidly Progressive or Atypical Rashes

Any rash that is:

- Rapidly spreading or evolving
- Unresponsive to standard therapy
- Associated with mucosal involvement
- Accompanied by blistering or skin detachment
- Painful rather than pruritic (suggesting deeper pathology)

...warrants urgent assessment and possible biopsy.

4. Pigmentary Changes

New or changing pigmented lesions — particularly those with asymmetry, irregular borders, color variation, diameter >6 mm, or evolving morphology — must be evaluated for **malignant melanoma** using the **ABCDE criteria**:

- Asymmetry
- Border irregularity
- Color change or heterogeneity
- Diameter >6 mm (pencil eraser size)
- Evolving shape or size

IMPORTANT

Non-blanching purpura is NEVER a benign finding — always investigate for vasculitis, sepsis, or hematological causes.

The ABCDE criteria should be applied to ALL pigmented lesions — early detection of melanoma dramatically improves prognosis.

Meningococcal septicemia can progress from scattered petechiae to widespread purpura fulminans within hours — this is a medical emergency requiring immediate IV antibiotics (e.g., IV ceftriaxone 2g stat).

OSCE PEARLS

Always perform diascopy when you see a red/purple rash — this distinguishes vasculitis from simple erythema.

Examine the palms, soles, nails, and oral mucosa — many systemic diseases show characteristic findings at these sites.

Document the morphology, distribution, and pattern of any skin lesion systematically.

Characteristic Skin Findings Indicating Underlying Systemic Disease

1. Toxic Erythema (Morbilliform Drug Eruption Viral Exanthem)

A widespread symmetrical **maculopapular erythema** is the most common drug eruption pattern. It begins on the trunk and spreads centrifugally to the limbs, is usually asymptomatic or mildly pruritic, and characteristically **blanches on diascopy** (key distinguishing feature from purpura/vasculitis).

Common drug culprits include amoxicillin, sulfonamides, allopurinol, carbamazepine, and phenytoin. Viral exanthems (e.g., measles, rubella, parvovirus B19, EBV) produce a similar morphological pattern.

2. Erythema Multiforme

Erythema multiforme presents as **target lesions** — characteristic round lesions with three concentric zones: a central dusky or blistering zone, a middle pale edematous ring, and an outer erythematous ring. It typically affects acral sites (hands, feet, extensor surfaces) symmetrically.

The most common trigger is **herpes simplex virus (HSV)** infection, though associations also include *Mycoplasma pneumoniae*, hepatitis B/C, and various drugs. Erythema multiforme is generally self-limiting but can recur with each HSV reactivation.

3. Erythema Nodosum

Erythema nodosum is the most common form of **panniculitis** (inflammation of subcutaneous fat). It presents as tender, erythematous, subcutaneous nodules, typically located on the **pretibial surfaces** (shins). The nodules are deep, indurated, and exquisitely tender to palpation.

Systemic associations include:

- Streptococcal infection (especially in children and young adults)
- Sarcoidosis
- Inflammatory bowel disease (IBD)
- Tuberculosis
- Behçet's disease
- Certain drugs (sulfonamides, oral contraceptive pill)

Diagnosis is usually clinical, though a deep skin biopsy showing septal panniculitis without vasculitis confirms the diagnosis. Management focuses on treating the underlying cause, bed rest, leg elevation, and NSAIDs for symptomatic relief.

4. Pyoderma Gangrenosum

Despite its name, pyoderma gangrenosum is **neither infectious nor gangrenous**. It is a neutrophilic dermatosis characterized by rapidly progressive, painful ulcers with **violaceous undermined borders** surrounding a necrotic base.

Systemic associations include:

- Inflammatory bowel disease (ulcerative colitis > Crohn's disease)
- Rheumatoid arthritis
- Hematological malignancy (myelodysplastic syndrome, acute myeloid leukemia)
- Hepatitis (particularly hepatitis C)

A key diagnostic feature is **pathergy** — the phenomenon whereby new lesions develop at sites of minor trauma (e.g., surgical wounds, IV catheter sites, venipuncture). This means that **debridement or surgical excision of pyoderma gangrenosum ulcers is contraindicated** and can cause worsening.

Treatment involves systemic immunosuppression — high-dose systemic corticosteroids (e.g., prednisolone 0.5–1 mg/kg/day) and/or cyclosporine are first-line agents.

5. Acanthosis Nigricans

Acanthosis nigricans presents as **symmetrical, velvety, hyperpigmented plaques** in intertriginous areas (axillae, groin, posterior neck, inframammary folds). The velvety texture is pathognomonic.

Associations include:

- **Insulin resistance and type 2 diabetes mellitus** (most common in obese patients)
- **Gastric adenocarcinoma** (when acanthosis nigricans is of sudden onset, rapidly progressive, and widespread — especially involving mucosal surfaces — this is termed "malignant acanthosis nigricans")
- Endocrine disorders (Cushing syndrome, polycystic ovary syndrome, acromegaly)

When acanthosis nigricans appears suddenly in a non-obese patient, urgent investigation for underlying malignancy is warranted.

6. Dermatomyositis (Gottron Papules and Heliotrope Rash)

Dermatomyositis is an idiopathic inflammatory myopathy with distinctive cutaneous manifestations:

- **Gottron papules:** Violaceous, flat-topped papules and plaques over the dorsal interphalangeal (IP) and metacarpophalangeal (MCP) joints. These are pathognomonic for dermatomyositis.
- **Heliotrope rash:** A violaceous or lilac-colored periorbital erythema, often associated with periorbital edema. Named after the heliotrope flower.
- **Mechanic's hands:** Hyperkeratotic, fissured, "dirty-appearing" skin on the lateral and palmar aspects of the fingers.

Dermatomyositis is associated with underlying malignancy in approximately 20–25% of adult patients. A thorough age-appropriate cancer screening should be performed in all adults with new-onset dermatomyositis, with particular attention to ovarian, lung, and gastrointestinal malignancies.

7. Systemic Lupus Erythematosus (SLE) Rashes

SLE produces several characteristic cutaneous manifestations:

- **Malar (butterfly) rash:** A fixed erythematous, edematous rash over the malar eminences and bridge of the nose, **sparing the nasolabial folds** (key distinguishing feature from other facial rashes). It is photosensitive.
- **Discoid lupus erythematosus:** Well-demarcated, erythematous, scaly plaques with follicular plugging, central atrophy, and dyspigmentation. Can cause permanent scarring alopecia when affecting the scalp.
- **Subacute cutaneous lupus erythematosus (SCLE):** Annular or psoriasiform photosensitive plaques, typically on the upper trunk and arms. Strongly associated with anti-Ro/SSA antibodies.

8. Necrolytic Migratory Erythema

Necrolytic migratory erythema is a characteristic rash associated with **glucagonoma** (a pancreatic neuroendocrine tumor secreting glucagon). It presents as:

- Erythematous, erosive, crusted, and scaling plaques
- Typically affects the groin, perineum, buttocks, and lower abdomen
- Often begins as a blistering eruption that evolves into erosive, crusted plaques
- Associated with glossitis, angular cheilitis, weight loss, and diabetes mellitus

The triad of necrolytic migratory erythema, diabetes mellitus, and weight loss in the context of a pancreatic mass is highly suggestive of glucagonoma.

KEY TEACHING POINTS

Erythema nodosum is the most common panniculitis — always look for underlying streptococcal infection, sarcoidosis, or IBD.

Pyoderma gangrenosum is a diagnosis of exclusion — it is NOT infectious and should NOT be debrided (pathergy phenomenon).

Malignant acanthosis nigricans (sudden onset, widespread, mucosal involvement) is a red flag for gastric adenocarcinoma.

Dermatomyositis in adults warrants thorough cancer screening — particularly ovarian, lung, and GI malignancies.

Necrolytic migratory erythema + diabetes + weight loss = think glucagonoma.

MCQ TRAPS & PITFALLS

Do NOT confuse blanching maculopapular rash (toxic erythema) with non-blanching palpable purpura (vasculitis).

Pyoderma gangrenosum is NOT caused by infection — do NOT prescribe antibiotics as primary treatment.

The malar rash of SLE spares the nasolabial folds — this distinguishes it from other facial rashes (e.g., seborrheic dermatitis, rosacea).

Erythema multiforme target lesions have THREE concentric zones — not two.

Characteristic Skin Findings and Their Systemic Associations

Skin Finding	Key Clinical Features	Systemic Associations
Toxic erythema (morbilliform)	Widespread symmetrical maculopapular erythema; blanches on diascopy	Drug reaction (amoxicillin, sulfonamides, allopurinol); viral exanthem
Erythema multiforme	Target lesions (3 concentric zones); acral distribution	HSV (most common trigger); Mycoplasma pneumoniae; Hepatitis B/C
Erythema nodosum	Tender, erythematous subcutaneous nodules on shins	Streptococcal infection; Sarcoidosis; IBD; TB; Behçet's disease
Pyoderma gangrenosum	Rapidly progressive painful ulcer with violaceous undermined borders	IBD (UC > CD); Rheumatoid arthritis; Hematological malignancy
Acanthosis nigricans	Symmetrical velvety hyperpigmented plaques in flexures	Insulin resistance/Type 2 DM; Gastric adenocarcinoma (malignant type)
Dermatomyositis	Gottron papules; Heliotrope rash; Mechanic's hands	Ovarian, lung, GI malignancy (20–25% of adult cases)
SLE rashes	Malar rash sparing nasolabial folds; Discoid plaques; SCLE	Systemic lupus erythematosus (multi-organ autoimmune disease)

Skin Finding	Key Clinical Features	Systemic Associations
Necrolytic migratory erythema	Erosive, crusted, scaling plaques in groin/perineum	Glucagonoma (pancreatic neuroendocrine tumor)

Nail and Hair Findings in Systemic Disease

Nail Changes

The nails can provide important diagnostic clues to systemic disease:

- **Clubbing (Hippocratic fingers):** Loss of the normal Lovibond angle (normally $\sim 160^\circ$), increased nail bed fluctuation, and a drumstick-like appearance of the distal phalanx. Associated with bronchogenic carcinoma, bronchiectasis, infective endocarditis, cyanotic congenital heart disease, IBD, and hepatic cirrhosis.
- **Beau's lines:** Transverse grooves in the nail plate caused by temporary arrest of nail matrix growth during a severe systemic illness (e.g., severe infection, myocardial infarction, chemotherapy). They appear weeks after the inciting event and grow out with the nail.
- **Onycholysis:** Separation of the nail plate from the nail bed, beginning distally and progressing proximally. Associated with psoriasis, thyroid disease (particularly hyperthyroidism), iron deficiency, and trauma.
- **Koonychia (spoon nails):** Concave, spoon-shaped nails. Classically associated with iron deficiency anemia, but also seen in hemochromatosis, thyroid disease, and Plummer-Vinson syndrome.
- **Muehrcke's lines:** Paired, white, transverse bands that do NOT move with nail growth (unlike Beau's lines). Associated with hypoalbuminemia (serum albumin < 2.2 g/dL), particularly in nephrotic syndrome and hepatic failure.
- **Splinter hemorrhages:** Small, linear, red-brown streaks in the nail bed. Associated with infective endocarditis, psoriasis, trauma, and vasculitis.
- **Half-and-half nails (Lindsay's nails):** The proximal half of the nail is white and the distal half is pink/brown. Associated with chronic renal failure (present in up to 40% of patients on dialysis).

Hair Changes

- **Alopecia areata:** Well-circumscribed, non-scarring patches of hair loss. An autoimmune condition associated with other autoimmune diseases (thyroid disease, vitiligo, pernicious anemia, type 1 diabetes mellitus).
- **Telogen effluvium:** Diffuse, non-scarring hair loss occurring 2–3 months after a significant physiological stressor (e.g., major surgery, severe illness, childbirth, crash dieting, iron deficiency). Caused by premature shift of hair follicles from anagen (growth) to telogen (resting) phase.
- **Hirsutism:** Excessive terminal hair growth in androgen-dependent areas in women (upper lip, chin, chest, lower abdomen). Associated with polycystic ovary syndrome (PCOS), Cushing syndrome, congenital adrenal hyperplasia, and androgen-secreting tumors.

- **Hypertrichosis:** Excessive hair growth in non-androgen-dependent areas. Can be congenital or acquired. Acquired hypertrichosis is associated with malignancy (paraneoplastic phenomenon), minoxidil, cyclosporine, and porphyria cutanea tarda.

IMPORTANT

Clubbing is the single most important nail sign — always investigate for underlying cardiorespiratory or gastrointestinal pathology.

Muehrcke's lines do NOT move with nail growth (unlike Beau's lines) — this is a key distinguishing feature.

Splinter hemorrhages in infective endocarditis are typically proximal (near the nail fold), while traumatic splinter hemorrhages are usually distal.

OSCE PEARLS

When you see clubbing, always check for hypertrophic pulmonary osteoarthropathy (HPOA) — look for wrist tenderness and swelling.

Examine all 20 nails — systemic diseases often affect multiple nails simultaneously.

Ask about the timeline of hair loss — telogen effluvium occurs 2–3 months after the precipitating event.

Nail Findings and Systemic Associations

Nail Finding	Description	Systemic Associations
Clubbing	Loss of Lovibond angle; drumstick appearance	Bronchogenic carcinoma; Bronchiectasis; Infective endocarditis; IBD
Beau's lines	Transverse grooves in nail plate	Severe systemic illness; Chemotherapy; MI
Onycholysis	Separation of nail plate from nail bed	Psoriasis; Hyperthyroidism; Iron deficiency
Koonychia	Spoon-shaped, concave nails	Iron deficiency anemia; Hemochromatosis; Thyroid disease
Muehrcke's lines	Paired white transverse bands (fixed)	Hypoalbuminemia; Nephrotic syndrome; Hepatic failure
Splinter hemorrhages	Linear red-brown streaks in nail bed	Infective endocarditis; Psoriasis; Vasculitis
Half-and-half nails	Proximal white, distal pink/brown	Chronic renal failure (up to 40% of dialysis patients)

Skin Manifestations of Specific Systemic Conditions

1. Diabetes Mellitus

Diabetes mellitus produces a wide range of cutaneous manifestations:

- **Necrobiosis lipoidica diabetorum:** Well-demarcated, yellowish-brown, atrophic plaques with a violaceous border, typically on the shins. The epidermis becomes so thin that dermal blood vessels are visible. Can ulcerate with trauma. Only ~0.3% of diabetic patients develop this, but when present, it is highly suggestive of diabetes.
- **Diabetic dermopathy (shin spots):** Brown, atrophic, macular lesions on the shins. The most common cutaneous finding in diabetes (present in up to 50% of diabetic patients). Often bilateral and asymptomatic.
- **Acanthosis nigricans:** As described above — velvety hyperpigmentation in flexures, indicating insulin resistance.
- **Granuloma annulare:** Erythematous or skin-colored annular (ring-shaped) plaques, typically on the dorsal hands and feet. The generalized form (widespread lesions) may be associated with diabetes mellitus.
- **Skin infections:** Diabetic patients are prone to candidal intertrigo, bacterial cellulitis, and mucormycosis (a life-threatening fungal infection of the rhinocerebral region in poorly controlled diabetics).

2. Thyroid Disease

- **Hyperthyroidism (Graves' disease):** Warm, moist, smooth skin; fine hair; onycholysis; pretibial myxedema (localized dermopathy — waxy, indurated, orange-peel-like plaques on the shins, specific to Graves' disease); and thyroid acropachy (digital clubbing with periosteal new bone formation).
- **Hypothyroidism:** Dry, coarse, cool skin; coarse, brittle hair; diffuse hair loss (including the outer third of the eyebrows — **Queen Anne's sign**); myxedema (non-pitting edema due to glycosaminoglycan deposition in the dermis); and carotenemia (yellowish discoloration of the skin, particularly palms and soles, due to elevated serum carotene).

3. Hepatic Disease

- **Jaundice:** Yellow discoloration of the skin and sclerae due to elevated serum bilirubin ($>35 \mu\text{mol/L}$). First visible in the sclerae and sublingual mucosa.
- **Spider naevi (spider angiomas):** Central arteriole with radiating capillary branches, typically on the drainage area of the superior vena cava (face, neck, upper trunk, arms). More than 5 spider naevi in a woman (not on OCP) or any number in a man is abnormal and suggests hepatic dysfunction.
- **Palmar erythema:** Diffuse reddening of the thenar and hypothenar eminences. Associated with hepatic cirrhosis, pregnancy, and rheumatoid arthritis.
- **Caput medusae:** Dilated periumbilical veins radiating outward from the umbilicus, indicating portal hypertension and portosystemic shunting.
- **Pruritus:** Generalized itching, particularly in cholestatic liver disease (primary biliary cholangitis), due to bile salt deposition in the skin.

4. Renal Disease

- **Uremic pruritus:** Generalized itching in chronic kidney disease, particularly in patients on hemodialysis.

- **Half-and-half nails (Lindsay's nails):** As described above.
- **Calciophylaxis (calcific uremic arteriolopathy):** A rare but life-threatening condition in end-stage renal disease, characterized by painful, violaceous, reticulated skin lesions (livedo reticularis-like) that progress to necrotic ulcers. Due to small vessel calcification and thrombosis.
- **Pericardial friction rub and uremic frost:** In severe uremia, urea crystals can deposit on the skin surface (uremic frost) — a rare finding in the modern era of dialysis.

5. Malignancy (Paraneoplastic Skin Manifestations)

Several skin conditions are recognized as paraneoplastic phenomena:

- **Acanthosis nigricans (malignant type):** Sudden onset, rapidly progressive, widespread — especially involving mucosal surfaces. Strongly associated with gastric adenocarcinoma.
- **Dermatomyositis:** As described above — 20–25% of adult cases are associated with underlying malignancy.
- **Erythema gyratum repens:** A rare, pathognomonic paraneoplastic rash characterized by concentric, wood-grain-like erythematous bands that migrate across the skin at a rate of up to 1 cm/day. Associated with lung, breast, and esophageal cancers in >80% of cases.
- **Leser-Trélat sign:** Sudden eruption of multiple seborrheic keratoses, often with associated pruritus. Associated with gastrointestinal adenocarcinoma and lymphoma.
- **Acquired ichthyosis:** Generalized dry, scaly skin resembling fish scales, developing in adulthood. Associated with Hodgkin lymphoma, mycosis fungoides, and other malignancies.
- **Sweet syndrome (acute febrile neutrophilic dermatosis):** Tender, erythematous plaques and nodules with a dense neutrophilic infiltrate on biopsy. Associated with hematological malignancies (particularly AML), solid tumors, and inflammatory conditions.

KEY TEACHING POINTS

Necrobiosis lipoidica is highly specific for diabetes, though it only occurs in ~0.3% of diabetic patients.

Pretibial myxedema is specific to Graves' disease (not hypothyroidism) — it is an autoimmune manifestation.

More than 5 spider naevi in a woman (not on OCP) or any number in a man is abnormal and suggests hepatic dysfunction.

Erythema gyratum repens is pathognomonic for underlying malignancy — if you see it, investigate aggressively.

The sudden eruption of multiple seborrheic keratoses (Leser-Trélat sign) should prompt investigation for GI malignancy.

MCQ TRAPS & PITFALLS

Pretibial myxedema occurs in HYPERthyroidism (Graves' disease), NOT hypothyroidism — do not confuse the two.

Diabetic dermopathy (shin spots) is the MOST common cutaneous finding in diabetes, not necrobiosis lipoidica.

Acquired ichthyosis (adult-onset) is a red flag for lymphoma — congenital ichthyosis is benign.

Spider naevi in the SVC drainage area (face, neck, upper trunk, arms) are abnormal — those on the lower body are usually benign.

Paraneoplastic Skin Manifestations and Associated Malignancies

Skin Manifestation	Key Features	Associated Malignancies
Malignant acanthosis nigricans	Sudden onset, widespread, mucosal involvement	Gastric adenocarcinoma
Dermatomyositis	Gottron papules, heliotrope rash	Ovarian, lung, GI cancers
Erythema gyratum repens	Concentric wood-grain-like migrating bands	Lung, breast, esophageal cancers (>80%)
Leser-Trélat sign	Sudden eruption of multiple seborrheic keratoses	GI adenocarcinoma, lymphoma
Acquired ichthyosis	Generalized dry, scaly skin in adulthood	Hodgkin lymphoma, mycosis fungoides
Sweet syndrome	Tender erythematous plaques, neutrophilic infiltrate	AML, solid tumors

Drug Reactions and the Skin

Severe Cutaneous Adverse Drug Reactions (SCARs)

Drug reactions are among the most important dermatological emergencies. Several severe cutaneous adverse drug reactions (SCARs) carry significant morbidity and mortality:

1. Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN)

SJS and TEN are considered part of the same spectrum of disease, differentiated by the extent of skin detachment:

- **SJS:** Skin detachment involving <10% of body surface area (BSA)
- **SJS/TEN overlap:** 10–30% BSA detachment
- **TEN:** >30% BSA detachment

Both conditions are characterized by widespread painful erythema, blistering, and full-thickness epidermal necrosis with detachment, resembling a scald burn. Mucosal involvement (oral, ocular, genital) is universal and often severe. Patients are at high risk of sepsis, fluid and electrolyte imbalance, and multi-organ failure.

Common drug culprits: Allopurinol, sulfonamides (including co-trimoxazole), carbamazepine, phenytoin, phenobarbital, lamotrigine, nevirapine, NSAIDs (oxicam type), and sertraline.

SCORTEN is a validated severity-of-illness score used to predict mortality in SJS/TEN. It assesses 7 independent risk factors: age >40, malignancy, heart rate >120, BSA detachment >10%, serum urea >10 mmol/L, serum bicarbonate <20 mmol/L, and serum glucose >14 mmol/L.

Mortality: SJS ~5–10%, TEN ~30–50%.

Management: Immediate withdrawal of the causative drug, supportive care in a burns unit or ICU, fluid and electrolyte management, wound care, ophthalmology review (to prevent ocular complications including symblepharon and corneal scarring), and consideration of IV immunoglobulin (IVIG) or cyclosporine.

2. Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS Syndrome)

DRESS syndrome is a severe, potentially fatal drug reaction characterized by:

- Widespread maculopapular rash (often progressing to exfoliative dermatitis)
- Fever
- Facial edema (a characteristic early feature)
- Lymphadenopathy
- Eosinophilia ($>1.5 \times 10^9/L$)
- Internal organ involvement: hepatitis (most common), nephritis, pneumonitis, myocarditis, thyroiditis
- Onset typically 2–8 weeks after starting the offending drug (later than most drug reactions)

Common drug culprits: Allopurinol, carbamazepine, phenytoin, sulfonamides, minocycline, abacavir, and vancomycin.

Management: Withdrawal of the causative drug, systemic corticosteroids (e.g., prednisolone 1 mg/kg/day with slow taper over 6–8 weeks to prevent relapse), and monitoring/treatment of organ involvement.

3. Acute Generalized Exanthematous Pustulosis (AGEP)

AGEP is characterized by the rapid onset of hundreds of small, non-follicular, sterile pustules on a background of widespread erythema, typically accompanied by fever and neutrophilia. It usually occurs within 24–48 hours of drug exposure and resolves spontaneously within 1–2 weeks after drug withdrawal.

Common drug culprits: Aminopenicillins, macrolides, calcium channel blockers, and antimalarials.

IMPORTANT

SJS/TEN and DRESS are medical emergencies — immediate drug withdrawal is the single most important intervention.

DRESS syndrome has a delayed onset (2–8 weeks) — always ask about drugs started up to 2 months before presentation.

AGEP pustules are STERILE — do not confuse with bacterial pustular infections.

Ocular complications of SJS/TEN can cause permanent blindness — early ophthalmology review is essential.

OSCE PEARLS

When assessing a drug reaction, always document the timeline of drug initiation relative to symptom onset.

Calculate SCORTEN within the first 24 hours of admission for SJS/TEN — it predicts mortality.

In DRESS syndrome, monitor LFTs, renal function, and cardiac enzymes — multi-organ involvement is common.

Severe Cutaneous Adverse Drug Reactions — Comparison

Feature	SJS/TEN	DRESS Syndrome	AGEP
Onset after drug	1–4 weeks	2–8 weeks	24–48 hours
Skin findings	Widespread erythema, blistering, epidermal detachment	Maculopapular rash → exfoliative dermatitis	Hundreds of small sterile pustules on erythema
Mucosal involvement	Universal (oral, ocular, genital)	Variable	Rare
Systemic features	Sepsis, fluid/electrolyte imbalance, multi-organ failure	Fever, eosinophilia, hepatitis, nephritis, myocarditis	Fever, neutrophilia
Key lab findings	Elevated SCORTEN score	Eosinophilia, atypical lymphocytes, elevated LFTs	Neutrophilia
Mortality	SJS 5–10%; TEN 30–50%	~10%	<1%
Management	Drug withdrawal; supportive care in burns unit/ICU; consider IVIG/cyclosporine	Drug withdrawal; systemic corticosteroids; organ monitoring	Drug withdrawal; self-resolving in 1–2 weeks

Skin Infections: When Infection Signals Systemic Disease

1. Cellulitis and Erysipelas

Cellulitis is an acute bacterial infection of the dermis and subcutaneous tissue, most commonly caused by **Streptococcus pyogenes** (Group A Streptococcus) and **Staphylococcus aureus**. It presents with localized erythema, warmth, swelling, and tenderness, often with systemic features (fever, malaise, raised inflammatory markers).

Erysipelas is a more superficial infection (upper dermis and superficial lymphatics) with a characteristically well-demarcated, raised border. It is almost exclusively caused by *Streptococcus pyogenes*.

Management: Flucloxacillin 500mg PO QID x 5–7 days (first-line for cellulitis); benzylpenicillin IV for severe cases or erysipelas. Elevation of the affected limb is important.

Complications: Necrotizing fasciitis (a surgical emergency — look for pain disproportionate to clinical findings, crepitus, skin necrosis, and systemic sepsis), recurrent cellulitis (consider prophylactic penicillin V 500mg OD in patients with ≥2 episodes per year).

2. Necrotizing Fasciitis

Necrotizing fasciitis is a rapidly progressive, life-threatening infection of the fascia and subcutaneous tissue that causes extensive necrosis. It is a **surgical emergency** requiring immediate debridement.

Key clinical features:

- Pain disproportionate to clinical findings (the hallmark)
- Rapidly spreading erythema and swelling

- Skin changes: dusky discoloration, bullae formation, crepitus (gas in tissues), skin necrosis
- Systemic toxicity: high fever, tachycardia, hypotension, confusion
- Elevated LRINEC score (Laboratory Risk Indicator for Necrotizing Fasciitis)

Management: Immediate surgical exploration and debridement, broad-spectrum IV antibiotics (e.g., meropenem + clindamycin + penicillin), and ICU support. Mortality is 20–40% even with treatment.

3. Disseminated Fungal Infections

In immunocompromised patients (HIV/AIDS, transplant recipients, chemotherapy patients), disseminated fungal infections can present with cutaneous manifestations:

- **Disseminated candidiasis:** Erythematous papules and nodules, often with a pale center (target-like appearance).
- **Cryptococcosis:** Umbilicated papules resembling molluscum contagiosum.
- **Histoplasmosis:** Oral ulcers, papules, nodules, and panniculitis.
- **Mucormycosis:** Black necrotic eschar (particularly rhinocerebral in diabetic patients).

4. Syphilis (Secondary)

Secondary syphilis presents with a widespread, non-pruritic, symmetrical rash that characteristically involves the **palms and soles**. The rash may be macular, papular, or papulosquamous. Other features include:

- Condylomata lata (moist, flat-topped papules in the perineum and intertriginous areas)
- Mucous patches (painless oral ulcers)
- Generalized lymphadenopathy
- Patchy alopecia ("moth-eaten" appearance)
- Fever, malaise, headache

Diagnosis is by serology: **VDRL/RPR** (non-treponemal screening test) and **TPHA/FTA-ABS** (treponemal confirmatory test). Treatment is with **intramuscular benzathine penicillin G 2.4 MU single dose**.

KEY TEACHING POINTS

Necrotizing fasciitis is a surgical emergency — pain disproportionate to clinical findings is the hallmark.

Recurrent cellulitis (≥ 2 episodes/year) warrants prophylactic penicillin V 500mg OD.

Secondary syphilis rash characteristically involves the palms and soles — always examine these sites.

Disseminated fungal infections in immunocompromised patients can mimic many other conditions — maintain a high index of suspicion.

MCQ TRAPS & PITFALLS

Erysipelas has a WELL-DEMARCATED raised border, while cellulitis has an ILL-DEFINED border — this is the key distinguishing feature.

Necrotizing fasciitis requires SURGICAL debridement, not just antibiotics — do not delay surgery.

Condylomata lata (syphilis) can be confused with genital warts (HPV) — remember that condylomata lata are MOIST and FLAT-TOPPED.

Skin Infections and Key Diagnostic Features

Infection	Causative Organism(s)	Key Clinical Features	First-Line Treatment
Cellulitis	Strep. pyogenes; S. aureus	Localized erythema, warmth, swelling, tenderness	Flucloxacillin 500mg PO QID
Erysipelas	Strep. pyogenes	Well-demarcated raised border; superficial	Benzylpenicillin IV or Penicillin V PO
Necrotizing fasciitis	Polymicrobial or Group A Strep	Pain disproportionate to findings; crepitus; necrosis	Emergency surgical debridement + broad-spectrum IV antibiotics
Secondary syphilis	Treponema pallidum	Non-pruritic rash on palms/soles; condylomata lata	IM benzathine penicillin G 2.4 MU single dose
Disseminated candidiasis	Candida species	Erythematous papules/nodules in immunocompromised	IV fluconazole or echinocandin

Psychodermatology: The Mind-Skin Interface

The relationship between psychological state and skin disease is bidirectional and complex. Psychological stress can trigger or exacerbate numerous skin conditions, and visible skin disease causes significant psychological distress.

Conditions with a Strong Psychological Component

- **Psoriasis:** Stress is a well-established trigger for flares. Psoriasis is associated with depression (prevalence ~30%), anxiety, and increased suicide risk. The hypothalamic-pituitary-adrenal (HPA) axis and sympathetic-adrenal-medullary (SAM) system are involved in stress-induced flares.
- **Atopic eczema:** Stress-induced flares are common. The itch-scratch cycle is perpetuated by anxiety and sleep deprivation.
- **Alopecia areata:** Strongly associated with psychological stress and psychiatric comorbidity (depression, anxiety, obsessive-compulsive disorder).
- **Chronic urticaria:** Stress can exacerbate symptoms. Up to 50% of patients with chronic spontaneous urticaria have associated psychological distress.
- **Delusions of parasitosis:** A monosymptomatic hypochondriacal psychosis in which patients have a fixed, false belief that their skin is infested with parasites. Patients often bring "specimens" (skin debris, fibers) in a matchbox — termed the **"matchbox sign"**. Treatment is with antipsychotic medications (e.g., risperidone, olanzapine).
- **Dermatitis artefacta (factitious dermatitis):** Patients intentionally produce skin lesions to assume the sick role. Lesions are often in accessible locations, have bizarre shapes inconsistent with any recognized dermatological condition, and the patient appears surprisingly unconcerned (**"la belle indifférence"**). Management involves gentle confrontation and psychiatric referral.

- **Trichotillomania:** Compulsive hair pulling, classified as an obsessive-compulsive spectrum disorder. Results in irregular patches of hair loss with hairs of varying lengths. Treatment includes cognitive behavioral therapy (CBT) and SSRIs.
- **Lichen simplex chronicus:** Localized area of lichenification (thickened skin with exaggerated skin markings) caused by chronic rubbing/scratching. Often associated with anxiety and stress.

Assessment Tools

- **DLQI (Dermatology Life Quality Index):** 10-item questionnaire, score 0–30, as described above.
- **Hospital Anxiety and Depression Scale (HADS):** Validated for use in dermatology patients.
- **Patient Health Questionnaire-9 (PHQ-9):** Screening tool for depression.

IMPORTANT

Always assess the psychological impact of skin disease — visible conditions carry profound psychosocial consequences.

Delusions of parasitosis requires antipsychotic medication, NOT antiparasitic treatment.

Dermatitis artefacta lesions are self-inflicted — direct confrontation is counterproductive; a gentle, non-judgmental approach is essential.

The "matchbox sign" (patient bringing skin debris in a container) is pathognomonic for delusions of parasitosis.

OSCE PEARLS

When assessing any skin condition, always ask: "How is this affecting your daily life, sleep, and mood?"

In dermatitis artefacta, the patient often appears surprisingly unconcerned about their lesions ("la belle indifférence").

For trichotillomania, look for hairs of varying lengths within the patch — this distinguishes it from alopecia areata.

Practice Questions

Q: A 45-year-old man presents with a 2-week history of painful, raised, purplish lesions on his lower legs that do not blanch on pressure. He also reports joint pain, fever, and weight loss. Urinalysis shows hematuria. What is the most likely diagnosis?

- A. Allergic contact dermatitis
- B. Erythema nodosum
- C. Leukocytoclastic vasculitis
- D. Cellulitis
- E. Psoriasis

Answer: C

Explanation: The key features pointing to leukocytoclastic vasculitis are: (1) non-blanching palpable purpura (indicating extravasation of red blood cells from inflamed vessels); (2) systemic symptoms (fever, weight loss, arthralgia); (3) end-organ involvement (hematuria indicating renal vasculitis). Erythema nodosum presents as tender subcutaneous nodules (not purpura), and cellulitis is typically unilateral with well-demarcated erythema.

Q: A 60-year-old woman presents with sudden onset of widespread, velvety, hyperpigmented plaques in her axillae and groin over the past 3 weeks. She has lost 8 kg of weight unintentionally. She is not obese. What is the most important investigation?

- A. Fasting glucose and HbA1c
- B. Upper gastrointestinal endoscopy
- C. Thyroid function tests
- D. Serum cortisol level
- E. Skin biopsy

Answer: B

Explanation: Sudden-onset, rapidly progressive acanthosis nigricans in a non-obese patient with unintentional weight loss is highly suggestive of malignant acanthosis nigricans, which is most strongly associated with gastric adenocarcinoma. Upper GI endoscopy is the most important investigation to identify the underlying malignancy. While insulin resistance (A) is the most common cause of acanthosis nigricans, the sudden onset and weight loss in a non-obese patient point toward a paraneoplastic etiology.

Q: A 32-year-old woman presents with violaceous, flat-topped papules over the dorsal interphalangeal joints, a lilac-colored periorbital rash, and hyperkeratated, fissured skin on the lateral aspects of her fingers. She reports progressive proximal muscle weakness. What is the most important next step?

- A. Prescribe high-potency topical corticosteroids
- B. Perform age-appropriate cancer screening
- C. Start oral antihistamines
- D. Refer for patch testing
- E. Prescribe oral aciclovir

Answer: B

Explanation: The clinical picture is classic for dermatomyositis: Gottron papules (violaceous papules over dorsal IP joints), heliotrope rash (lilac periorbital erythema), mechanic's hands (hyperkeratotic fissured skin on lateral fingers), and proximal myopathy. In adults, dermatomyositis is associated with underlying malignancy in 20–25% of cases, particularly ovarian, lung, and gastrointestinal cancers. Age-appropriate cancer screening is essential. Topical steroids (A) would not address the systemic disease, and the other options are not relevant to this presentation.

Q: A 28-year-old man presents with a widespread, non-pruritic, symmetrical rash involving the palms and soles. He also has generalized lymphadenopathy, patchy hair loss, and moist, flat-topped papules in the perineum. What is the most appropriate diagnostic test?

- A. Skin biopsy

- B. VDRL/RPR and TPHA/FTA-ABS
- C. Blood cultures
- D. KOH preparation of skin scrapings
- E. Patch testing

Answer: B

Explanation: The presentation is classic for secondary syphilis: widespread non-pruritic rash involving the palms and soles, condylomata lata (moist flat-topped perineal papules), generalized lymphadenopathy, and "moth-eaten" alopecia. Diagnosis is confirmed by serological testing: VDRL/RPR (non-treponemal screening) and TPHA/FTA-ABS (treponemal confirmatory). Treatment is IM benzathine penicillin G 2.4 MU single dose.

Q: A 55-year-old woman presents with a rapidly progressive, painful ulcer on her left lower leg. The ulcer has violaceous, undermined borders and a necrotic base. She has a history of ulcerative colitis. What is the most important aspect of management?

- A. Surgical debridement of the ulcer
- B. Broad-spectrum IV antibiotics
- C. Systemic corticosteroids
- D. Compression bandaging
- E. Skin grafting

Answer: C

Explanation: The clinical presentation is classic for pyoderma gangrenosum: rapidly progressive painful ulcer with violaceous undermined borders in a patient with ulcerative colitis. The key management principle is that pyoderma gangrenosum exhibits pathergy — new lesions develop at sites of trauma. Therefore, surgical debridement (A) and skin grafting (E) are contraindicated and can cause worsening. Broad-spectrum antibiotics (B) are not effective as this is not an infectious condition. First-line treatment is systemic immunosuppression with corticosteroids (e.g., prednisolone 0.5–1 mg/kg/day) and/or cyclosporine.

Q: A 40-year-old man presents with a 3-day history of widespread painful erythema, blistering, and skin peeling affecting 35% of his body surface area. He has severe oral and ocular mucosal involvement. He was started on allopurinol 4 weeks ago for gout. His SCORTEN score is 4. What is the most likely diagnosis?

- A. Erythema multiforme major
- B. Stevens-Johnson syndrome
- C. Toxic epidermal necrolysis
- D. DRESS syndrome
- E. Staphylococcal scalded skin syndrome

Answer: C

Explanation: Toxic epidermal necrolysis (TEN) is defined as skin detachment involving >30% of body surface area. This patient has 35% BSA involvement, placing him in the TEN category. The timeline (4 weeks after starting allopurinol), widespread painful erythema, blistering, full-thickness epidermal necrosis, and severe mucosal involvement are all consistent with TEN. Allopurinol is one of the most common drug culprits. A SCORTEN of 4 corresponds to a predicted mortality of ~55%. SJS (B) involves <10% BSA detachment. DRESS (D) typically presents with a maculopapular rash and eosinophilia, not full-thickness epidermal necrosis. SSSS (E) is caused by Staphylococcal exfoliative toxins and typically affects children.

Q: A 35-year-old woman presents with tender, erythematous, subcutaneous nodules on both shins. She reports a sore throat 2 weeks ago. She has no history of inflammatory bowel disease or sarcoidosis. What is the most likely underlying cause?

- A. Tuberculosis
- B. Streptococcal infection
- C. Behçet's disease
- D. Sarcoidosis
- E. Oral contraceptive pill use

Answer: B

Explanation: Erythema nodosum is the most common form of panniculitis, presenting as tender, erythematous subcutaneous nodules on the pretibial surfaces. In children and young adults, the most common cause is streptococcal infection. The history of a sore throat 2 weeks prior strongly suggests a post-streptococcal etiology. While TB (A), sarcoidosis (D), Behçet's disease (C), and OCP use (E) are all recognized associations, streptococcal infection is the most likely cause in this clinical context.

Q: A 50-year-old man with chronic kidney disease on hemodialysis presents with painful, violaceous, reticulated skin lesions on both thighs that are progressing to necrotic ulcers. What is the most likely diagnosis?

- A. Calciphylaxis (calcific uremic arteriopathy)
- B. Necrotizing fasciitis
- C. Leukocytoclastic vasculitis
- D. Warfarin-induced skin necrosis
- E. Pyoderma gangrenosum

Answer: A

Explanation: Calciphylaxis (calcific uremic arteriopathy) is a rare but life-threatening condition seen in patients with end-stage renal disease, particularly those on hemodialysis. It is characterized by painful, violaceous, reticulated skin lesions (livedo reticularis-like) that progress to necrotic ulcers, due to small vessel calcification and thrombosis. The clinical context (hemodialysis patient) and the description of the lesions are classic for calciphylaxis. Management includes wound care, sodium thiosulfate, and optimization of calcium-phosphate metabolism.

Q: A 25-year-old woman presents with irregular patches of hair loss on her scalp. On examination, the patches contain hairs of varying lengths. She reports that she has been pulling out her hair when stressed. What is the most likely diagnosis?

- A. Alopecia areata

- B. Tinea capitis
- C. Trichotillomania
- D. Telogen effluvium
- E. Traction alopecia

Answer: C

Explanation: Trichotillomania is a compulsive hair-pulling disorder classified as an obsessive-compulsive spectrum disorder. The hallmark finding is irregular patches of hair loss with hairs of varying lengths (broken at different lengths), which distinguishes it from alopecia areata (where the patches are smooth with no hairs within them). The patient's own report of pulling hair when stressed confirms the diagnosis. Treatment includes cognitive behavioral therapy (CBT) and SSRIs. Alopecia areata (A) presents with smooth, completely bald patches and exclamation mark hairs at the periphery.

Q: A 65-year-old man presents with a 6-week history of a widespread, intensely pruritic rash consisting of erythematous plaques and nodules. He has a history of treated acute myeloid leukemia. Biopsy shows a dense dermal neutrophilic infiltrate without evidence of infection. What is the most likely diagnosis?

- A. Cutaneous T-cell lymphoma
- B. Sweet syndrome (acute febrile neutrophilic dermatosis)
- C. Leukocytoclastic vasculitis
- D. Erythema multiforme
- E. Drug eruption

Answer: B

Explanation: Sweet syndrome (acute febrile neutrophilic dermatosis) is characterized by tender, erythematous plaques and nodules with a dense neutrophilic infiltrate on biopsy. It is strongly associated with hematological malignancies, particularly acute myeloid leukemia (AML). The clinical context (history of AML), the morphology of the lesions (erythematous plaques and nodules), and the biopsy findings (dense neutrophilic infiltrate without infection) are all consistent with Sweet syndrome. Treatment is with systemic corticosteroids.